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Unit 1 - Exploring the Core

PDF Documents

PDF Document 1: 1738068886773-Chapter 4 - Unit 1 - Exploring the Core.pdf

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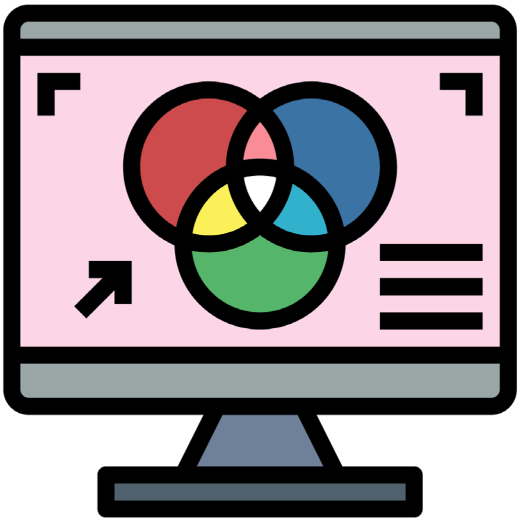
UNIT 1 EXPLORING THE CORE

Topic Investigation



UNDERSTANDING THE SCIENCE OF COLOR MIXING

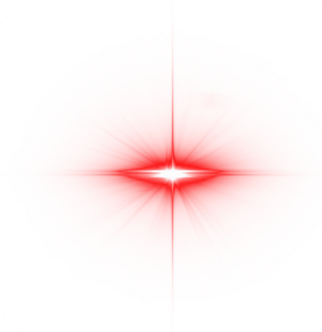
Color plays a pivotal role in how we perceive the world around us, and nowhere is this more evident than in the way we use and interact with light. In this section, we delve into the science behind the primary colors of light—red, green, and blue—and their ability to combine to create a broad spectrum of colors. This phenomenon is known as additive color mixing and is fundamental to various technologies you use every day, like televisions, smartphones, and computer screens.

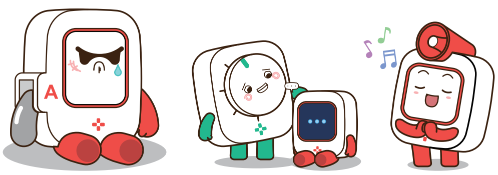
THE BASICS OF ADDITIVE COLOR MIXING

Additive color mixing occurs when light colors combine to produce a new color by the process of light addition. This method uses light to mix colors, with red, green, and blue being the primary colors. When these colors are mixed in different combinations and intensities, they can produce a wide range of colors, including secondary colors like cyan, magenta, and yellow.

RED LIGHT

This color has the longest wavelength visible to the human eye and is often associated with warmth and urgency. It's powerful for grabbing attention and is frequently used in warning signs and to signal danger.



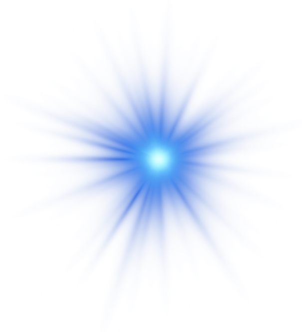


GREEN LIGHT

Located in the middle of the light spectrum, green light is the easiest color for our eyes to process. It is often used to represent safety and permission, which is why traffic lights use green to signify 'go'.

BLUE LIGHT

With a shorter wavelength, blue light is calming and is commonly used in environments that require relaxation and calmness. It's also the color of light most pervasive in digital screens, which is why looking at your phone or computer at night can sometimes make it hard to sleep.

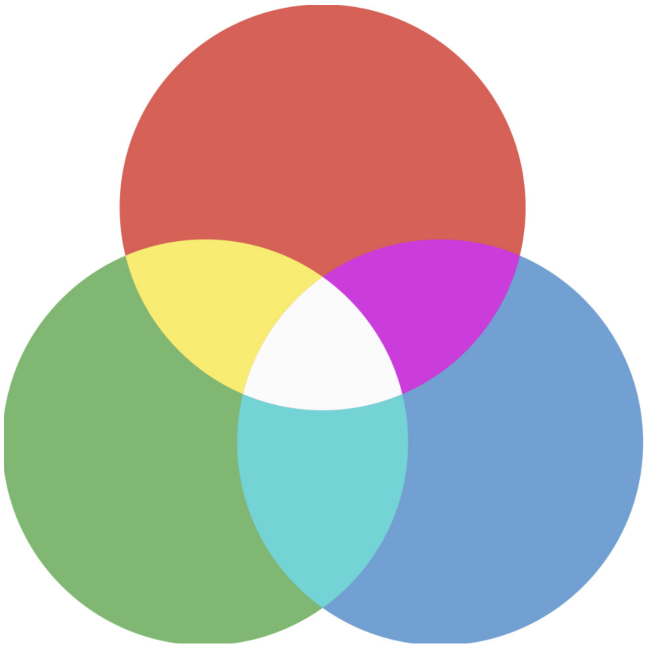


EXPERIMENTING WITH LIGHT

To see how these colors mix, you can perform a simple light mixing experiment:

1. SETUP

In a dark room, use three flashlights, each covered with either red, green, or blue transparent materials. These will act as your primary light sources.



2. OBSERVATION

Shine each flashlight onto a white wall. Start with one color, then gradually introduce the others.

3. COLOR MIXING

Observe the colors that result from the overlaps:

- Red and green lights mix to produce yellow.
- Green and blue lights combine to form cyan.
- Blue and red lights create magenta.
- All three colors together will result in white.

Inquiry & Reflection

Discussion question 1

How do colors influence your mood or actions in daily life?
Discuss examples of where colors play a crucial role in daily decision-making, such as in traffic signals or in choosing what to wear based on the colors that make you feel confident or relaxed.

Discussion question 2

Why do you think different devices have different screen color settings, like 'Night Mode'?
Explore how the color of light emitted by device screens might affect your eyes or sleep patterns, and why it might be beneficial to change these settings based on the time of day.